

Milan Kumar Modak

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Computer Science undergraduate specializing in AI/ML and full-stack development, with hands-on experience building machine learning, computer vision, and LLM-powered applications. Experienced with Python, FastAPI, React, TypeScript, SQL, and open-source development.

Education

- Bachelor of Technology Computer Science and Engineering | O.P. Jindal University | Raigarh, C.G**
 - 8.4 GPA overall as of 6 semesters2023 – 2027
- Senior Secondary | St. Michael's Eng. Med. School | Raigarh, C.G.**
 - Percentage : 75.8%Passing Year : 2023
- High School | St. Michael's Eng. Med. School | Raigarh, C.G.**
 - Percentage : 70%Passing Year : 2021

Certification

- Data Analytics Essentials**
 - Issued By : CISCOIssued : Dec 2024
- Introduction to Internet of Things (Elite)**
 - Issued By : NPTELIssued : Dec 2025

Skills

Languages: Python, Java, C, SQL

ML/Data: Machine Learning, Scikit-learn, Pandas, Data Science, Computer Vision, NLP

Web: React, Node.js, FastAPI, HTML, CSS, Tailwind CSS

Tools/Platforms: Git, GitHub, Linux, MongoDB

AI: LLMs, Groq API, YOLOv8, EasyOCR, Tesseract

Experience

JUNE 2026 – AUGUST 2026

Open Source Contributor | Vicharanashala (VLED Lab) | IIT Ropar, Punjab

- Fixed a leaderboard performance bug in ViBe, an open-source ed-tech learning platform built with React, Node/Express, and MongoDB; changes merged via PR #1236.
- Onboarded onto a large-scale TypeScript/pnpm monorepo with Firebase authentication, resolving environment setup and integration issues.
- Collaborated with a team building a friction-detection feature for Shamagama using DBSCAN clustering and LLM-based alerts

AUG 2025 – SEPT 2025

Project Intern – Saiket Systems

➤ Customer Churn Prediction: –

- Built binary classification models using XGBoost and Random Forest to predict customer churn.
- Performed EDA and feature engineering on 5,000+ customer records and addressed class imbalance using SMOTE.
- Analyzed feature importance to identify factors associated with customer churn and support retention strategies.

Projects

1. **Phishing Detection Extension:** Established a browser extension, to detect and block phishing websites in real-time using URL analysis, blacklist checks, and content inspection, enhancing user security during web browsing.
2. **Automated Number Plate Detection (ANPR):** Built an end-to-end ANPR (Automatic Number Plate Recognition) pipeline combining **YOLOv8 for plate detection** and **EasyOCR/Tesseract for text extraction** from vehicle images/video frames. Implemented image preprocessing techniques using **OpenCV** (grayscale, thresholding, noise reduction) to improve OCR accuracy on varied lighting and angle conditions
3. **GitHub Repo Assistant**
 - Built a full-stack web app that helps developers quickly understand unfamiliar GitHub repositories, using React, TypeScript, and Tailwind CSS on the frontend and FastAPI on the backend.
 - Integrated the Groq API (LLM) to automatically generate repo summaries, setup instructions, and codebase walkthroughs from a given GitHub URL.
 - Deployed frontend on Vercel and backend on Render, handling CORS, environment configuration, and API integration for a production-ready deployment. Tech: React, TypeScript, Tailwind CSS, FastAPI, Python, Groq API, Vercel, Render.

Hobbies

- Cricket • Exercise • Online Gaming